NAME: _____ Geometry (FL B.E.S.T.)

8.5 Dilations: Perimeter and Area

A dilation with scale factor k centered at the origin sends point (x, y) to (kx, ky) — every length in the figure is multiplied by $|k|$. Since perimeter is a sum of lengths, it scales the same way: $P' = |k| \cdot P$. Area, however, is a *product* of two lengths (like base times height), so it scales by k^2: $A' = k^2 \cdot A$. This squared relationship also works in reverse — if you know how areas compare, take a square root to find how sides or perimeters compare, and vice versa. This worksheet includes coordinate dilations and real-world scale-model problems that combine both relationships.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Perimeter 14; $k = 4$. Image perimeter?

2. Area 10; $k = 3$. Image area?

3. Perimeter 24 maps to 8. Find k.

4. Area 50; $k = \frac{1}{5}$. Image area?

5. Sides in ratio $2 : 5$. Area ratio?

6. Areas in ratio $9 : 1$. Perimeter ratio?

7. Perimeter 36; $k = \frac{2}{3}$. Image?

8. Area 18; $k = 2$. Image?

9. Areas 16 and 64 (original to image). Find k.

10. Side 8; $k = 1.5$. Image side?

11. Image area 45 after $k = 3$. Find the original area.

12. Perimeter 100; $k = 0.25$. Image?

NAME: _____ Geometry (FL B.E.S.T.)

13. Corresponding sides 4 and 6. Area ratio (first to second)?

14. A triangle with vertices $(0, 0)$, $(6, 0)$, $(0, 8)$ is dilated from the origin by $k = 1.5$. Find the perimeter and area of the image.

15. A poster's area increases from 24 in^2 to 96 in^2 under a dilation. If the original perimeter was 20 in, find the scale factor and the new perimeter.

✓ Answer Key

1. 56
2. 90
3. $\frac{1}{3}$
4. 2
5. 4 : 25
6. 3 : 1
7. 24
8. 72
9. 2
10. 12
11. 5
12. 25
13. 4 : 9
14. Perimeter = 36; area = 54
15. $k = 2$; new perimeter = 40 in